



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH
INDUSTRIAL & PRODUCTION ENGINEERING
4th SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Course Structure
(From Academic Session 2024-25 onwards)

B. Tech 4th Semester: Industrial & Production Engineering
Semester IV

Sl. No.	Course Code	Course Type	Course Title	Hours per Week			Credit	Marks	
				L	T	P	C	CE	ESE
Theory									
1	IP241401	PCC	Basic Machining Processes	3	0	0	3	30	70
2	ME241402	PCC	Dynamics of Machines	3	0	0	3	30	70
3	ME241403	PCC	Machine Drawing	2	0	2	3	30	70
4	ME241404	PCC	Material Science and Engineering	3	0	0	3	30	70
5	IP241405	PCC	Engineering Inspection and Metrology	3	0	0	3	30	70
6	HS241406	HSMC	Finance and Accounting	3	0	0	3	30	70
7	AU241407	AU	Environmental Science	3	0	0	0 (PP/NP)	100	-
Practical									
8	IP241411	PCC	Basic Machining Processes Lab	0	0	2	1	15	35
9	ME241412	PCC	Dynamics of Machine Lab	0	0	2	1	15	35
10	ME241414	PCC	Material Science and Engineering Lab	0	0	2	1	15	35
11	PR241415	PR	Micro Project (Skill Based)	0	0	4	2	15	35
TOTAL				20	0	12	23	340	560
Total Contact Hours per week: 32									
Total Credit: 23									

Course Code	Course Title	Hours per week L-T-P	Credit
IP241401	Basic Manufacturing Processes	3-0-0	3

Prerequisites: NIL

Course Outcome (CO): After completion of the course, the students will be able to

CO1: Develop knowledge on cutting tools, tool geometry and interpret mechanics of metal cutting.

CO2: Understand the constructional features, mechanism and working principles of the basic machining processes, work holding devices.

CO3: Analysis mathematically and determine the material removal rate and other machining parameters of the basic machining processes.

CO4: Develop knowledge of grinding and finishing processes.

MODULE 1:

Principal of metal cutting: Introduction to metal cutting. Concept of Material removal rate (MRR). Role of cutting tool and cutting parameters. Single and multipoint cutting tools.

Tool Geometry: Basic shape of cutting tool, Tool-in-hand nomenclature, ASA, ORS system of nomenclature, Inter-relationship between different systems.

Wear in cutting tools: Causes of wear machinability and tool life, Taylor's tool life equation and its calculation. Effect of various parameters on tool life.

Tool materials: Requirements/ properties. Carbon tool steels, HSS, Cast alloy steel, cemented carbides, ceramics, CBN, UCON, Stellite. **(4 Lectures)**

MODULE 2:

Mechanics of chip formation: Overview of chip formation, classification of chips, Built up edge (BUE). **Chip thickness ratio, velocity relationships.** Orthogonal cutting, oblique cutting, concept of shear plane, shear angle. Merchant's theory of metal cutting and Force relationship; stress and strain and power consumption. Heat zones metal cutting. Calculation of shear angle, chip thickness ratio. **(5 Lectures)**

MODULE 3:

Lathe: Engine Lathe– block diagram– sketch– functions of each part– work holding devices in lathe– functions Chuck, Centre, Dogs, Steady Rest and Follower Rest, mechanism of lathe– Apron, Feed, Tumbler Gear, various operations performed in Lathe– facing, turning, chamfering and knurling– relative positions of tool and job– Taper turning operations. Method of thread cutting– selection and arrangement of tool and work. Specifications of Lathe. Calculation of taper angle, machining time, Material removal rate (MMR). Capstan and turret Lathe; principles and working. **(6 Lectures)**

MODULE 4:

Drilling, Boring and Reaming: Purpose of drilling– block diagram and function– types of drilling machines– portable drilling– bench type– sensitive drilling– radial arm drilling– functions of parts– purpose and operation– gang milling, multiple drill head, upright drilling, relative operations– reaming, boring, tapping, counter boring, courses sinking, trepanning and spot facing (with simple sketch, purpose and application). Work holding devices– specification torque calculation– speed, feed and depth of cut. Numerical **(6 Lectures)**

MODULE 5:

Shaping, Planning, Slotting and Broaching Machine:

Purpose of shaping– block diagram– functions of each part. Purpose of planer– block diagram– functions of each part. Purpose of slotting machine– block diagram– functions and working principle. Operations carried out– horizontal plane, vertical plane, Comparison of planer with shaper– work

holding devices in shaper and planer– Quick return mechanism in shaper-mechanical and hydraulic– cross feed mechanism– Stroke length and Position adjustment of the ram- Types of planer with application– mechanism in planer– Comparison of shaping with slotting– tool holding devices in shaper, planer and slotter-specifications of shaper, planer and slotter simple problems to calculate the velocity– speed, feed and depth of cut. Broaching Machine: Purpose, tool and machine. **(7 Lectures)**

MODULE 6:

Milling: Milling machine purpose– up and down milling– classification of milling machines– Horizontal milling machine– block diagram– functions of each part applications– Vertical milling machine– block diagram– functions of each part applications– Gear cutting using milling machine– procedure with neat sketch– milling cutters– peripheral, face, end T slot, form etc.– attachments and special accessories for milling– rotary, slotting attachment– indexing mechanism– methods of indexing– direct, plain, 6compound and differential indexing– problems– specifications– cutting conditions and parameters. **(5 Lectures)**

MODULE 7:

Grinding and Finishing: Types, specification and size of grinding machine. Grinding wheel – abrasive, bond, grit, grain, structure; wheel dressing, truing, balancing and mounting selection of wheels. Operation –grinding of flat, cylindrical and contoured surfaces. Centreless grinding. Different finishing processes –Grinding, Lapping, Honing and super-finishing –Polishing, Buffing, Tumbling and Burnishing. **(3 Lectures)**

Text Books/References:

1. A. Bhattacharyya, Metal cutting theory and practice, Pub- New Central Book Agency (P) Ltd. Kolkata, Latest Edition
2. R.K.Jain, Production Technology, Khanna Publishers, New Delhi, 2001/Latest Edition
3. Hajra Choudhary et. al., Elements of Production Technology–Vol.II, Asia Publishing House, 2010/Latest Edition
4. B. Kumar, Manufacturing Technology, Khanna Publishers, New Delhi, 2000/Latest Edition
5. Stephenson & Agapion, Metal cutting theory and practice, Taylor and Francis(group) – NY., Latest Edition

Course Code	Course Title	Hours per week L-T-P	Credit
ME241402	Dynamics of Machines	3-0-0	3

Prerequisites: Theory of Machines

Course Outcome (CO): After completion of the course, the students will be able to

CO1: Analyze velocity and acceleration in mechanisms by utilizing kinematic principles.

CO2: Assess the performance of governors in controlling speed fluctuations and maintaining stability in mechanical systems.

CO3: Evaluate stability and control in rotating systems by using gyroscopic principles.

CO4: Apply balancing techniques for rotating masses to balance several masses rotating in same plane and different planes.

CO5: Apply balancing techniques for reciprocating masses to minimize vibrations, improve stability, and enhance the efficiency of mechanical systems and engines.

MODULE 1: Analysis & Synthesis of Mechanism (8 Lectures)

Relative velocity method for finding velocity and acceleration in link, slider crank mechanism, and four bar chain; Coriolis acceleration component, rubbing velocity at pin joint, forces acting in a mechanism, mechanical advantage, Davis and Ackermann steering gear, universal Hook joint, synthesis of mechanism, analysis vs. synthesis of mechanism, number and dimensional synthesis, function generation by graphical method for two position and three position synthesis, Freudenstein's equation, Chebyshev spacing method for precision points.

MODULE 2: Governors & their performance (8 Lectures)

Function and types of governors, Watt, Porter, and Proell governor; Hartnell, Wilson-Hartnell, and Hartung spring-controlled governors, sensitiveness, stability, isochronism, hunting, effort & power, controlling force, stability of spring-controlled governor, friction and insensitiveness.

MODULE 3: Gyroscopic Effects & Applications (8 Lectures)

Concept of spinning and precession, gyroscopic couple, effect of gyroscopic couple on rotor on a shaft, airplane, naval ship, four-wheeler automotive vehicle, and two-wheeler vehicle; gyroscope.

MODULE 4: Balancing of Rotating Masses (8 Lectures)

Need of balancing, balancing of single rotating mass in same plane and in different plane, analytical and graphical methods for balancing of several masses rotating in same plane and rotating in different planes.

MODULE 5: Balancing of Reciprocating Masses (10 Lectures)

Balancing of reciprocating engine, partial balancing of primary force, partial balancing of locomotives, variation of tractive force, swaying couple, hammer blow, primary balance of multi cylinder in-line engine, secondary balance of multi cylinder in-line engine, V-engine balancing, method of direct and reverse cranks.

Text Books/References:

1. Dr. R.K. Bansal, Dr. J.S. Brar, Theory of Machines, Laxmi Publications(P) Ltd, 5th edition, 2016
2. Ratan S.S., Theory of Machines, McGraw Hill Education, 4th edition, 2017
3. Prof. P. L. Ballaney, Theory of Machines and Mechanisms, Khanna Publishers, 25th edition, 2014
4. Sadhu Singh, Theory of Machines, Pearson Education India, 3rd edition 2011
5. John J. Uicker, Jr., Gordon R. Pennock and Joseph E. Shigley, Theory of Machines and Mechanisms, Oxford University Press, 5th edition 2017

Course Code	Course Title	Hours per week L-T-P	Credit
ME241403	Machine Drawing	2-0-2	3

Prerequisites:

Course Outcome (CO): After completion of the course, the students will be able to

CO1: Illustrate the Orthographic and Sectional views of machine components.

CO2: Draw the profiles of thread and locking devices such as nuts and bolts.

CO3: Sketch various joints-key and cotter joints, riveted joints, welded joints, coupling, pipe joints as well as machine elements.

CO4: Assemble the components of engine and valve parts such as piston, stuffing box, cross head, connecting rod, eccentric, feed check valve Illustrate, stop valve, blow off cock, non-return valve.

CO5: Understand the symbols and notations used in the production drawing.

MODULE 1: Orthographic and Sectional views (6 Lectures)

Introduction, 1st and 3rd angle projection, Front, Top and Side views of a machine component, Sectional views, Half-sectional views of machine components.

MODULE 2: Screw Fasteners (6 Lectures)

Introduction, Screw Thread Nomenclature, Forms of Thread, Thread Profiles- V-Thread, Buttress Thread, British Standard Whitworth (B.S.W) Thread, Square Thread ACME Thread, Worm Thread, Thread designation, Multi-start Thread, left hand thread, right hand thread, Locking devices for nuts, Different types of bolts and nuts.

MODULE 3: Keys, Joints and couplings (8 Lectures)

Introduction, Saddle Key, Sunk Key, Cotter Joint with Sleeve, Socket and Spigot Ends, and with Gibs, Knuckle joints. Rigid Coupling Flanged Coupling, Sleeve or Muff Coupling, Flexible Coupling, Pipe Joints, Hydraulic Joints.

MODULE 4: Riveted and Welded Joints (6 Lectures)

Introduction, Rivet heads, Definitions-Pitch, Margin, Chain Riveting, Zig-Zag Riveting, welded joints and Symbols, Dimensioning of welds.

MODULE 5: Assembly Drawings (10 Lectures)

a) Engine Parts: Stuffing Box, Crosshead, Connecting Rod, Eccentric, Piston

b) Valves: Stop Valve, Feed Check Valve, blow off Cock, Non-Return Valve

MODULE 6: Production Drawing (6 Lectures)

Introduction, Method and placing limit dimensions, Datum symbols, Tolerance of forms and positions, Surface roughness machining symbols, Direction of lay symbols on production or working drawing.

Text Books/References:

1. N.D Bhatt, Machine Drawing, Charotar Publishing House Pvt. Ltd., 51st edition 2022
2. K. L. Narayana, P. Kanniah, K. Venketa Reddy, Machine Drawing, New Age International Pvt. Ltd, 5th edition, 2016
3. N. Sidheswar, P. Kanniah and V.V.S. Sastry, Machine Drawing, McGraw Hill Education, 3rd edition 2017
4. K. L. Narayana, Production Drawing, New Age International Pvt. Ltd, 3rd edition 2014

Course Code	Course Title	Hours per week L-T-P	Credit
ME241404	Material Science and Engineering	3-0-0	3

Prerequisites: NIL

Course Outcome (CO): After completion of the course, the students will be able to

CO1: Analyse the failure of metals by studying the mechanical properties and different failure mechanism like fracture, fatigue and creep.

CO2: Identify crystal structures and imperfections for various materials.

CO3: Analyse binary phase diagrams of alloys including Fe-Fe₃C system

CO4: Design proper heat treatment process to tailor material properties for ferrous alloys.

CO5: Classify different Engineering alloys, namely Copper, Aluminium, Cast iron, stainless steel and non-metallic materials.

MODULE 1: Mechanical Property measurement (8 Lectures)

Tensile, compression and torsion tests, relations between true and engineering stress-strain curves, yielding and yield strength, ductility, resilience, toughness, Hardness: Rockwell, Brinell and Vickers and their relation to strength

Fracture of Metals: Ductile fracture, Brittle fracture, Ductile and Impact Testing, Toughness and Impact testing

Fatigue of Metals: Cyclic stress, SN curve, endurance and fatigue limits, Fatigue life calculations

Creep and Stress rupture of metals: Creep of metals, Creep test.

MODULE 2: Crystal Structures and Imperfection in solids (8 Lectures)

Crystal Structures: Crystal systems and Bravais Lattice, Principal Metallic Crystal Structure (SC, BCC, FCC, HCP), Miller Indices and direction for Crystallographic Planes in Cubic unit cell, Crystal structure analysis by X ray diffraction

Imperfection in solids: Point, line, interfacial and volume defects; deformation by twinning, Stacking faults.

MODULE 3: Phases and Phase Diagram (8 Lectures)

Phase diagram of pure substance, Gibbs Phase rule, Cooling curves, Lever rule, Binary phase diagram: Eutectic, Eutectoid, Peritectic, Peritectoid; Iron-Carbon equilibrium diagram, Microstructure changes during phase transformation.

MODULE 4: Heat Treatment of Plain Carbon Steel (6 Lectures)

Transformation in plain carbon steel: TTT diagram, Effect of Alloying elements in Alloy Steel, Different heat treatment processes: Hardening, Tempering, Annealing, Normalizing, Case Hardening, Cyaniding, Nitriding, Flame and Induction Hardening.

MODULE 5: Engineering Alloys (6 Lectures)

Cast iron: General Properties, Types of cast iron, Stainless steel: Ferritic, Martensitic and Austenitic stainless steel, Copper Alloy: General properties of copper, Production of Copper, Classification of Copper Alloy, Wrought Copper alloy, Aluminium Alloy: General properties of Aluminium and its production, Wrought Aluminium Alloys, Aluminium Casting Alloys.

MODULE 6: Non-Metallic Materials (6 Lectures)

Classification of non-metallic materials, Rubber: classification, Properties, processing and applications, Plastics: Thermosetting and Thermoplastics, General purpose Thermoplastics (Polyethylene, Polypropylene, Polystyrene), Engineering thermoplastics (Nylons, Polycarbonate, Acetals), Ceramics: Properties and applications, Processing of Ceramics, Glasses: Glass transition temperature, Structure and composition of glasses, Composites: Properties and applications, Fibers for Reinforced Plastic Composite Materials (Glass fibers, carbon fibers).

Text Books/References:

1. William D. Callister Jr. and David G. Rethwisch, Materials Science and Engineering: An Introduction, Wiley, 9th Edition 2013
2. James F. Shackelford, Introduction to Materials Science for Engineers, Pearson, 8th Edition, 2020
3. William F. Smith and Javad Hashemi, Foundations of Materials Science and Engineering, McGraw-Hill Education, 2009
4. S.L Kakani and Amit Kakani, Material Science, New Age International Private Limited, 2016
5. V. Raghavan, Materials Science and Engineering, Prentice Hall India Learning Private Limited, 6th Edition 2015

Course Code	Course Title	Hours per week L-T-P	Credit
IP241405	Engineering Inspection and Metrology	3-0-0	3

Course Outcomes (CO): At the completion of the course the student will be able:

CO1: To apply proper instruments for dimensional measurements according to level of precision.

CO2: To apply necessary standards for obtaining desired fit and design appropriate limit gauges.

CO3: To develop analytical and experimental techniques for precise measurement of thread and gear parameters.

CO4: To design appropriate control charts for statistical process control.

CO5: To estimate texture of machined surface by stylus equipment and optical interferometry.

MODULE 1: Introductory Concept (2 Lectures)

- Meaning of engineering metrology. Precision, Accuracy, Sources of errors in measurement.
- Meaning of engineering inspection, types of inspection, merit/demerit of 100% inspection.
- Sampling inspection – Representative sample. Standards of measurement and sub-division of standards.

MODULE 2: Statistical Process Control (6 Lectures)

- Dimensional variations during manufacture, Chance causes and Assignable causes.
- Control chart and its significance in statistical process control, Meaning of process under statistical control. Examples of control charts.
- Computer implementation of control charts.

MODULE 3: Tolerance, Limits of Size and Fits (6 Lectures)

- Concept of tolerance, allowance and clearance
- Natural tolerance limits, Process capability and Specification limits.
- Hole and shaft basis systems of specifying limits of size and tolerances.
- Indian Standard for fits and tolerances.
- Limit gauges – Meaning of gauge, Taylor's principle of limit gauging, Design of hole/ring gauge and plug gauge and their use, Type I and Type II statistical errors.
- Interchangeability – its importance in production, techniques of achieving interchangeability during manufacture of industrial products.
- Comparators – Features of comparators, classification of comparators, different comparators and their uses in mass production.

MODULE 4: Tool Room Measuring Instruments (6 Lectures)

Vernier Calliper, Micrometer screw gauge, Height gauge, Depth gauge, V blocks, Straight edges, Radius gauge, Feeler gauge, Wire gauge, Thread pitch gauge, Bevel protractor, Combination set, Bore gauge, Sine bar and slip gauges, Dial indicator with magnetic base, Surface plate, Profile projector (PP), Tool maker's microscope (TMM), Diameter measuring machine (DMM) and Coordinate measuring machine (CMM): Types of CMM, Role of CMM, and applications of CMM.

MODULE 5: Measurement of Screw Threads (5 Lectures)

- Parameters for measurement of screw threads.
- Measurement of various parameters of screw thread such as diameter, thread angle, effective diameter and pitch.
- Use of screw thread micrometer and Thread pitch gauge
- One wire, Two wire and Three wire methods.
- Use of TMM, PP, DMM in thread measurement.

MODULE 6: Measurement of Gears (5 Lectures)

- Profile of gear tooth and Involute function.

- (b) Spur gear measurements by functional and analytical tests. Parkson gear tester.
- (c) Measurement of tooth thickness – chordal thickness method, constant chord method, base tangent method.
- (d) Check for pitch circle diameter and tooth spacing.

MODULE 7: Surface Texture (4 Lectures)

- (a) Meaning of surface texture, Elements of surface texture.
- (b) Meaning of roughness and Waviness.
- (c) Roughness width cut-off (Sampling length) and its significance
- (d) Representation of surface roughness.
- (e) Procedure of estimation of surface roughness.

Measurement of surface roughness by stylus equipment.

MODULE 8: Interferometry (3 Lectures)

- (a) Condition for constructive and destructive interference of monochromatic light waves.
- (b) Sources of monochromatic light for lab use.
- (c) Use of optical flat.

Principle of Gauge length interferometer and Laser interferometer.

MODULE 9: Alignment Testing (3 Lectures)

- (a) Optical methods for alignment testing
- (b) Laser alignment testing.

Alignment tests on machine tools

Textbooks:

1. Metrology and Measurement, Anand Bewoor, Vinay A. Kulkarni, TMH
2. Engineering metrology – M Mahajan
3. Engineering Metrology – R K Jain
4. Dimensional metrology – M K Khare and S Vajpayee, OXFORD-IBH Publishers

Reference Books:

1. Handbook of industrial metrology – ASTM publication
2. Engineering Metrology – K J Hume, Published by Macdonald & Co.(1968)
3. Practical Engineering Metrology – K W B Sharp, Sir Isaac Pitman & Sons
4. Engineering Precision measurements – A W Judge, Chapman and Hall publishing (1957)
5. Dimensional Metrology – L Miller, Edward Arnold publishing Co
6. Precision Measurements – Jack Johnson – Pitman publishing Co.

Course Code	Course Title	Hours per week L-T-P	Credit
HS241406	Finance and Accounting	3-0-0	3

Prerequisites: High School Education

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Develop a strong foundation in accounting principles, including the classification of accounts, the Double Entry System, the application of the Golden Rules of Debit and Credit, record financial transactions in journals, post them into ledgers, and prepare a Trial Balance.
- **CO2:** Understand the role of subsidiary books in accounting, including different types of cash books and their preparation; learn to reconcile differences between the Cash Book and Pass Book balances and prepare a Bank Reconciliation Statement
- **CO3:** Differentiate between capital and revenue expenditure, understand the concepts of bad debts and provisions for doubtful debts and gain the skills to prepare final accounts by incorporating necessary adjustments.
- **CO4:** Explore the concepts of savings and investment, including their benefits and differences, identify various investment avenues in physical and financial assets and understand the distinction between trading and investment to make informed financial decisions.
- **CO5:** Have a comprehensive understanding of financial markets, covering money and capital markets, key financial instruments, IPOs, stock exchanges, and mutual funds, and analyze the structure and functions of financial markets and assess different financial instruments used for investment.
- **CO6:** Acquire knowledge and skills in stock market investment and analysis, applying fundamental and technical analysis to evaluate stocks, learn to interpret stock market charts and patterns, assess stock price movements, and utilize various analytical tools for informed investment decisions.

MODULE 1: Introduction to Accounting

Concept and classification of Accounts, Transaction, Double Entry System of Book Keeping, Golden rules of Debit and Credit, Journal- Definition, Advantages, Procedure of Journalising, Ledger, Advantages, Rules regarding Posting, Balancing of ledger accounts, Trial Balance- Definition, Objectives, Procedure of preparation.

MODULE 2: Subsidiary Books, Cash Book and Bank Reconciliation

Name of Subsidiary Books, Cash Book- Definition, Advantages, Objectives, Types of Cash Book, Preparation of different types of Cash Books, Bank Reconciliation Statement, Reasons of disagreement between Cash Book with Pass Book balance, Preparation of bank Reconciliation Statement.

MODULE 3: Expenditures, Bad Debts and Final Accounts

Concept of Capital Expenditure and Revenue Expenditure, Bad Debts, Provision for Bad and Doubtful debts, Final Accounts: Preparation of Trading Account, Profit and Loss account and Balance Sheet with adjustments.

MODULE 4: Introduction to Saving and Investment

Savings: Concept, Benefits and saving alternatives, Disadvantage of Saving – Time Value of Money. Investment: Meaning, Advantage, Investment Avenues – Investment in Physical and financial assets, Difference between Saving and Investment; Trading- Meaning, Difference between Trading and Investment.

MODULE 5: Financial Market and Instruments.

Financial Markets and Instruments: Money Market, Capital Market, Money Market Instruments and features. Capital Markets and Features, New Issue Market and Stock Market; New Issue market- Initial Public Offering , Market Players, Secondary Market :Equity and Debentures, BSE and NSE, Index and its uses; Mutual Funds: Meaning and Types.

MODULE 6: Stock Market Investment and Analysis

Stock Market Investment: Fundamental Analysis- Economic Analysis-Industry Analysis and Technical Analysis - Tools of technical analysis, Understanding various types of charts and patterns- (Line chart, Bar Chart, Candlestick Chart Point and Figure Chart)

Textbooks/Reference Books

1. Dam, B. B.Sarda, R.A. Barman, R. Kalita, B., Theory and Practice of Accountancy- Vol. I, Gayatri Publication, 2019
2. Goel, D.K Goel, R. Goel,S. Accountancy, Avichal publishing company, 2020
3. Kevin, S., Security Analysis and Portfolio Management, Prentice Hall India., 2022
4. Pathak,B., Indian Financial System, Pearson Education, 2024
5. Chandra, P., The Invesment Game: How To win, Tata McGrawHill, 2012

Course Code	Course Title	Hours per week L-T-P	Credit
AU241407	Environmental Science	3-0-0	0

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Relate environment and its associated problems
- **CO2:** Explain basic ethics of living and protection of other organisms
- **CO3:** Applying sustainable development principles

MODULE 1: Introduction

Environment and Ecology, Objectives of ecological study, Aspects of Ecology, Autecology, Synecology; Ecosystem, Structural and functional attributes of an ecosystem, Food chain and foodweb, Energy flow, Bio geochemical cycles

MODULE 2: Land: Use and Abuse

Land use: Impact of land–use on environmental quality, Land degradation, Control of land degradation, Wasteland, Wet lands

MODULE 3: Water Pollution

Introduction: Water quality standards, Water pollution

MODULE 4: Air Pollution

Introduction, Air pollution system, Air pollutants, Air pollution laws

MODULE 5: Noise Pollution

Sources of noise pollution, Effects of noise, Physical effects, Physiological effects

MODULE 6: Control of Pollutions

Control of water pollution and management, Water pollution legislations, Water quality management in surface water; Control of air pollution, source correction method, Pollution control equipment; controls of Noise pollution

Textbooks/Reference Books

1. Dr Suresh Dhameja, Environmental engineering and management, 2010
2. Dr B.S. Chauhan, Environmental studies
3. Henry and Hence, Environmental science and engineering
4. Dr Susmitha Baskar, Environmental studies for undergraduate course
5. Clair Sawyer, Chemistry for environmental engineering and science

Course Code	Course Title	Hours per week L-T-P	Credit
IP241411	Basic Machining Processes Lab	0-0-2	1

Prerequisites: NIL

Course Outcomes: At the completion of the practical, students will be able

- **CO1:** To use proper metal cutting tools and fixtures for producing desired parts.
- **CO2:** To apply various workshop machines for production of parts according to job design.
- **CO3:** To evaluate the effect of machining parameters on quality of machined components.

Exp. No. 1

Facing and Plain Turning

AIM

To perform the facing and plain turning in a centre lathe on the given cylindrical work piece for the given dimensions.

MATERIALS REQUIRED

Mild steel rod of diameter 25 mm and length 105 mm

TOOLS REQUIRED

Single point cutting tool, Vernier calliper, chuck key & tool post key

PROCEDURE

1. The dimensions of the given cylindrical work piece are checked
2. The work piece is held in the chuck properly and tightened by chuck key
3. The single point cutting tool is held on tool post and tightened by tool post key
4. The facing operation is done on both side of the work piece to get the required length
5. The plain turning operation is done on work piece to get the initial dimension
6. Then the plain turning operation is further continued to specific dimension to form steps
7. Finally, the dimensions are checked using Vernier calliper before work piece is removed from the chuck.



BEFORE MACHINING



**AFTER
MACHINING**

All dimensions are in mm

Exp. No. 2

Step Turning and Chamfering

AIM

To perform the Step Turning and Chamfering in a centre lathe on the given cylindrical work piece for the given dimensions.

MATERIALS REQUIRED

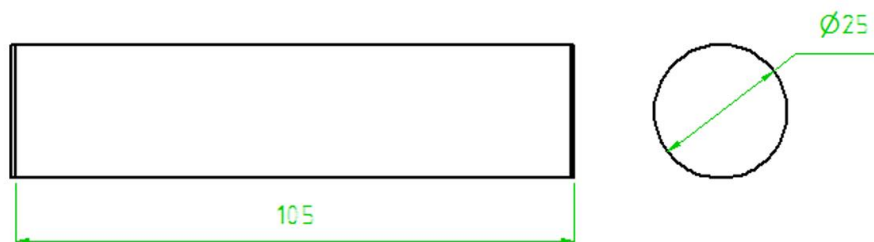
Mild steel rod of diameter 25 mm and length 105 mm

TOOLS REQUIRED

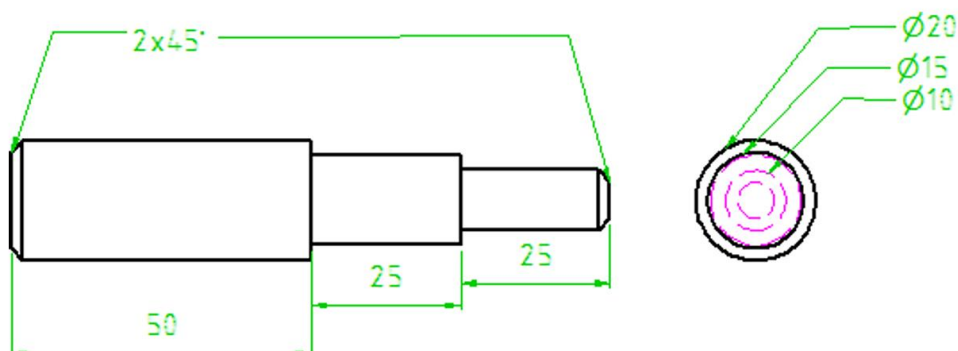
Single point cutting tool, Chamfer tool, Vernier calliper, and chuck key & tool post key

PROCEDURE

1. The dimensions of the given cylindrical work piece are checked
2. The work piece is held in the chuck properly and tightened by chuck key
3. The single point cutting tool is held on tool post and tightened by tool post key
4. The facing operation is done on both side of the work piece to get the required length
5. The plain turning operation is done on work piece to get the initial dimension
6. Then the plain turning operation is further continued to specific dimension to form steps
7. Thus, the step turning operation is done on the given work piece
8. Then using Chamfer tool, the chamfering operation is done on the workpiece.
9. Finally, the dimensions are checked using Vernier calliper before work piece is removed from the chuck.



BEFORE MACHINING



AFTER MACHINING

All dimensions are in mm

Exp. No. 3**Taper Turning****AIM**

To perform the taper turning using compound rest in a centre lathe on the given cylindrical work piece for the given dimensions.

MATERIALS REQUIRED

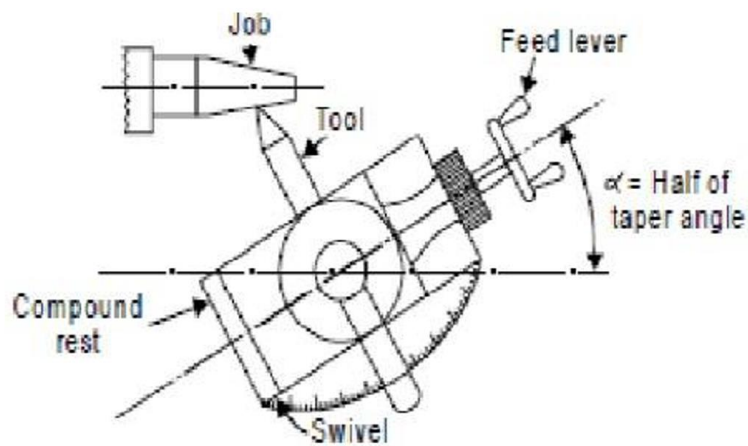
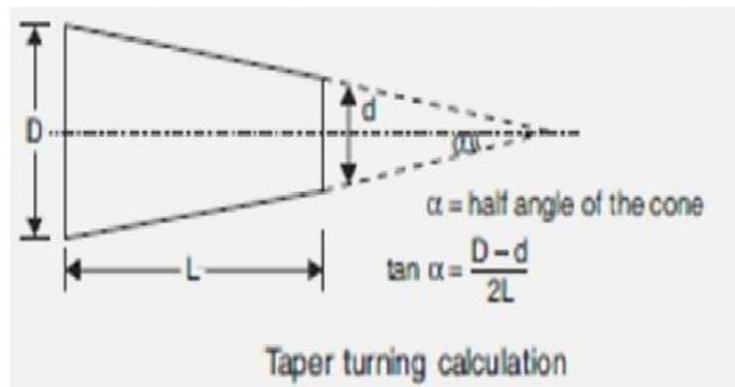
Mild steel rod of diameter 25 mm and length 105 mm

TOOLS REQUIRED

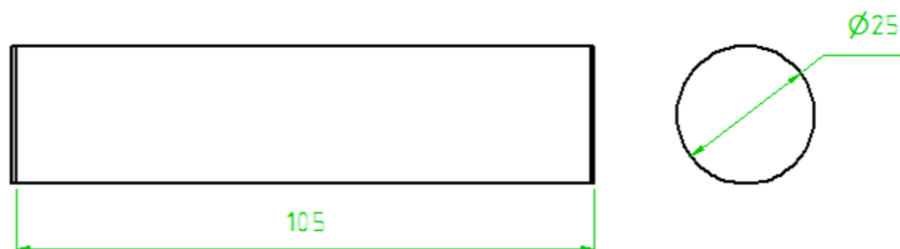
Single point cutting tool, Vernier calliper, chuck key & tool post key

PROCEDURE

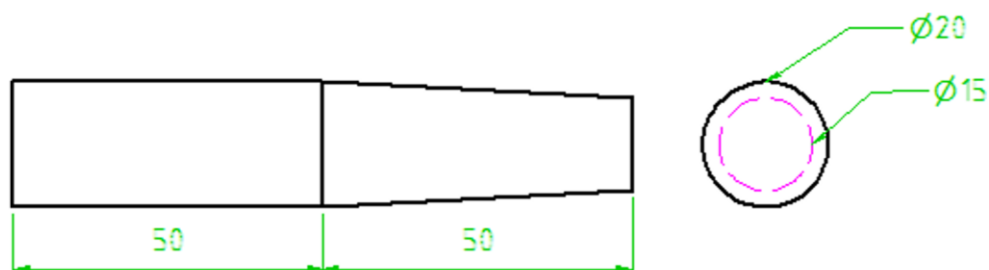
1. The dimensions of the given cylindrical work piece are checked
2. The work piece is held in the chuck properly and tightened by chuck key
3. The single point cutting tool is held on tool post and tightened by tool post key
4. The facing operation is done on both side of the work piece to get the required length
5. The plain turning operation is done on work piece to get the initial dimension
6. Then the plain turning operation is further continued to specific dimension to form steps
7. Thus, the step turning operation is done on the given work piece
8. Then the compound rest is adjusted to an angle found out by the formula to get the taper
9. position. $\tan \theta = D - d / 2L$, where D – large diameter; d – smaller diameter; L – length of the taper.
10. Thus, the compound rest is adjusted and saddle is moved for performing the operations.
11. Thus, the final taper shape is obtained and checked for dimensions before it is removed from the chuck.



Taper turning by swiveling compound rest



BEFORE MACHINING



AFTER MACHINING

All dimensions are in mm

Exp. No. 4

External Threads Cutting

AIM

To perform the thread cutting operations in a centre lathe on the given cylindrical work piece for the given dimensions.

MATERIALS REQUIRED

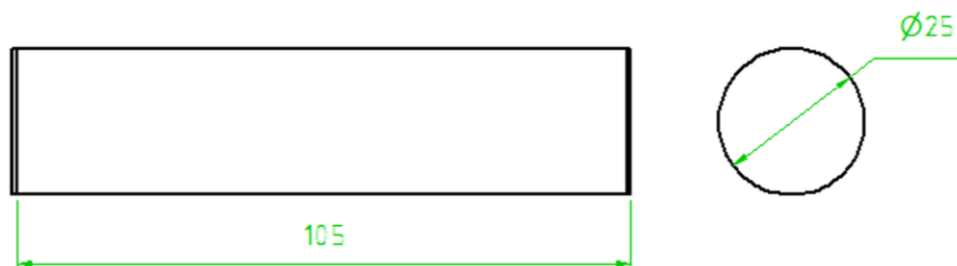
Mild steel rod of diameter 25 mm and length 105 mm

TOOLS REQUIRED

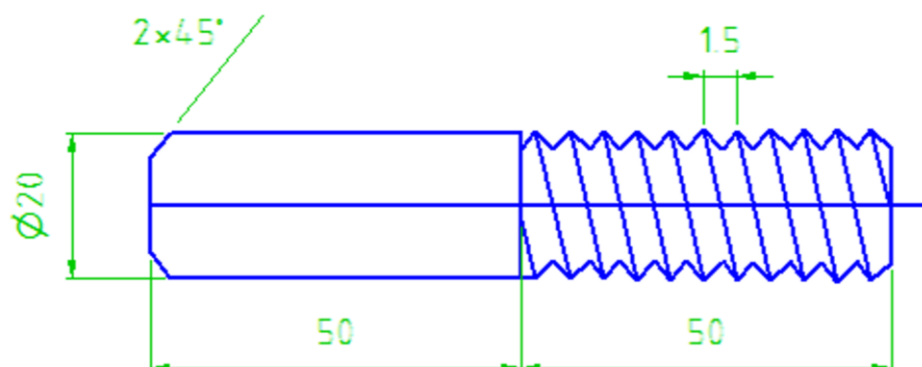
Single point cutting tool, Vernier calliper, chuck key & tool post key

PROCEDURE

1. The dimensions of the given cylindrical work piece are checked
2. The work piece is held in the chuck properly and tightened by chuck key
3. The single point cutting tool is held on tool post and tightened by tool post key
4. The facing operation is done on both side of the work piece to get the required length
5. The plain turning operation is done on work piece to get the initial dimension
6. Then the plain turning operation is further continued to specific dimension to form a step
7. Thus, the step turning operation is done on the given work piece
8. Then the lead screw is adjusted to automatic position for thread cutting operation such that the required depth of cut is given by cross slide movement.
9. Finally, the dimensions are checked using Vernier calliper before work piece is removed from the chuck.



BEFORE MACHINING



AFTER MACHINING

All dimensions are in mm

Exp. No. 5

Drilling, Boring and Reaming

AIM

To perform the drilling, Boring and Reaming in a drilling Machine on the given work piece for the given dimensions.

MATERIALS REQUIRED

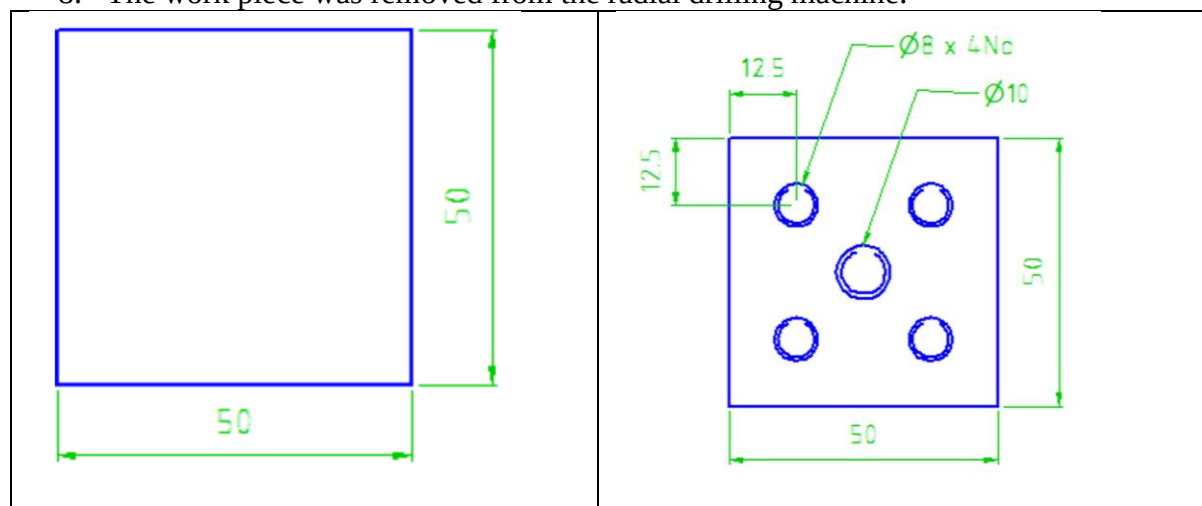
Mild steel Plate of Length 50x50 mm

TOOLS REQUIRED

Steel rule, Flat file (rough and smooth), Drill bit (8 mm, 10 mm, 10.5 mm), Reaming tool, Try square

PROCEDURE

1. The work piece was fitted in the vice and filed to the required dimensions.
2. The squareness of the work piece was checked.
3. Drawing punches were made for various drills.
4. The job was fitted on the radial drilling machine.
5. The 10 mm & 8 mm drill bit were used for drilling in the required place and drilling operation was made on the work piece.
6. Boring was done on the 10mm hole using the boring tool.
7. Reaming was done on the 8 mm hole using the Reaming tool size of 8 mm diameter.
8. The work piece was removed from the radial drilling machine.



All dimensions are in mm

Exp. No. 6

Indexing on milling machine

Aim:

To machine a hexagonal nut using indexing head on milling machine

MATERIALS REQUIRED

Mild steel Plate of Length 15 mm thickness

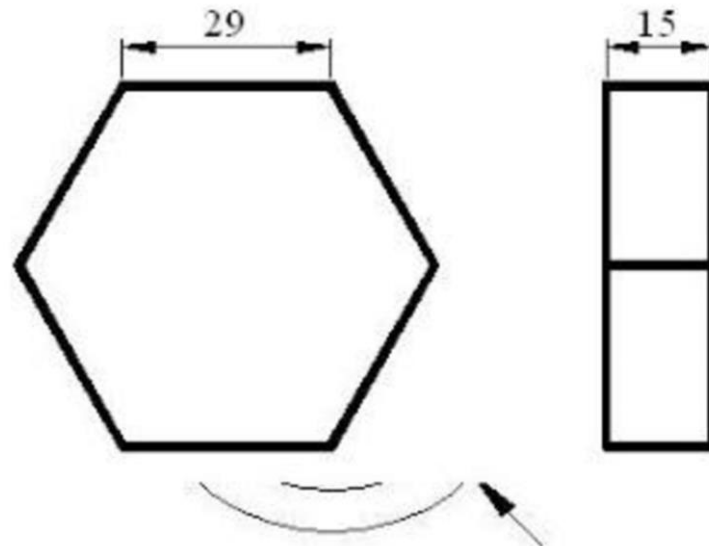
TOOLS REQUIRED:

Milling Machine, Indexing head, Divider, Steel Rule.

PROCEDURE:

1. Fit the helical cutter on the arbor and hold the specimen between the centers of the dividing head and the tail center with help of dog carrier.
2. Carefully adjust the work piece so that the cutter just touches the top surface of the specimen.

3. Calculate the necessary depth of cut and then mill the six faces of the hexagonal nut in succession.
4. C The workpiece is now checked for final dimensions.
5. The workpiece is now checked for final dimensions.



All dimensions are in mm

Exp. No. 7

Groove cutting on shaper machine

Aim:

To produce a rectangular groove on a given casted rectangular blank with the help of shaping operations.

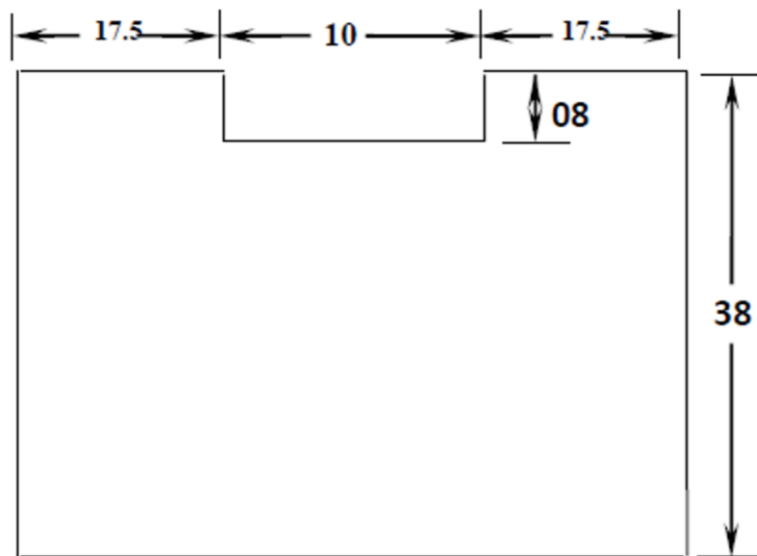
TOOLS REQUIRED:

Shaper machine, spirit level gauge, shaper tools, surface plate, vernier height gauge, scriber, vernier calliper.

PROCEDURE

- 1) Select suitable horizontal shaper.
- 2) Fix the given work piece on shaper table with the help of universal vice rigidly.
- 3) Set the cutting stroke and return stroke of the shaper, the length of the shaper should not exceed 3 times length of rectangular blank.
- 4) Select proper shaper tool and fix the tool on tool head of the shaper with minimum clearance of 5 to 6mm.
- 5) The exact position of the tool is perpendicular to the rectangular face of the work piece.
- 6) Switch on the shaper; feed the reciprocating tool against the rigidly fixed rectangular bank by table feeding mechanism.
- 7) Required depth of cut 1.5mm is given by rotating the tool head hand wheel with graduated micrometer.
- 8) Desired feed like 0.5mm/stroke is given to the work piece through the feeding mechanism incorporated for each cutting stroke up a length of 10mm.
- 9) Machine surface" A' de clamp and deburr corners.

- 10) Butt Finish the surface A against the fixed jaw so that A and B are perpendicular to each other.
- 11) Machine surface de clamp and deburr corners.
- 12) Check the squareness between surface A & B with tray square, if it is not square check the setting machine till squareness is obtained.
- 13) Butt surface B against the fixed jaw and rest surface A on the parallel and clamp.
- 14) Machine surface D to size and deburr check the size and squareness.
- 15) Machine surface E to size and check the squareness
- 16) Machine surface F to size and check the squareness.



Rectangular Groove on work piece

All dimensions are in mm

Course Code	Course Title	Hours per week L-T-P	Credit
ME241412	Dynamics of Machine Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

CO1: Compute Coriolis component of acceleration

CO2: Determine characteristics curves of for Watt, Porter, Proell and Hartnell governor for sleeve position against speed and radius of rotation against controlling force.

CO3: Determine percentage variation of theoretical and experimental gyroscopic couple

CO4: Determine Static Balance of a set of four blocks with unbalance by properly positioning them in angular and lateral position on the shaft.

CO5: Plot the $x - \theta$ (follower displacement vs. angle of cam rotation) curves for different cam follower pairs.

Module No.	Description	Contact hours
1	Coriolis Component of Acceleration Apparatus	4
2	Characteristics curves of for Watt, Porter, Proell and Hartnell governor for sleeve position against speed and radius of rotation against controlling force	4
3	Percentage variation of theoretical and experimental gyroscopic couple	4
4	Static Balance of a set of four blocks with unbalance by properly positioning them in angular and lateral position on the shaft.	4
5	Study the various cam and follower pairs, effect of follower weight on bounce the effect of spring compression on bounce	4
	Total Contact Hours	20

Text Books/References:

1. Dr. R.K. Bansal, Dr. J.S. Brar, Theory of Machines, Laxmi Publications(P) Ltd., 5th edition, 2016
2. Ratan S.S., Theory of Machines, McGraw Hill Education, 4th edition, 2017
3. Prof. P. L. Ballaney, Theory of Machines and Mechanisms, Khanna Publishers, 25th edition, 2014

Course Code	Course Title	Hours per week L-T-P	Credit
ME241414	Material Science and Engineering Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Compute the hardness of a flat mild steel and high carbon steel specimen
- **CO2:** Compute the impact resistance of a notched bar flexure specimen
- **CO3:** Compute the compressive strength of materials
- **CO4:** Conduct typical engineering microscopic observation to recognize the microstructure of cast iron metal.

Module No.	Description	Contact hours
1	Hardness Test To determine hardness of a flat mild steel and high carbon steel specimen by using Rockwell and Brinell Hardness Tester Machine.	4
2	Charpy Pendulum Impact Test To determine impact resistance of an assigned specimen in the form of a notched bar flexure specimen.	4
3	Compressive Strength Test To determine the compressive Strength of a given specimen.	4
4	Preparation of metallic specimen for metallographic study by grinding, polishing and etching.	2
5	Study of the Microstructures of Cast Iron	2
	<u>Total Contact Hours</u>	16

Text Books/References:

1. William F. Smith and Javad Hashemi, Foundations of Materials Science and Engineering, McGraw-Hill Education, 2009
2. S.L Kakani and Amit Kakani, Material Science, New Age International Private Limited, 2016

Course Code	Course Title	Hours per week L-T-P	Credit
PR241415	Micro Project (Skill Based)	0-0-4	2

Prerequisites: Knowledge of basic science

Objectives:

- To identify problems based on societal or research needs
- To acquaint with the process of solving the problem in a group
- To apply engineering knowledge to attempt solutions to the problems

Course Outcome (CO): After completion of the course, the students will be able to

- CO1: Analyze engineering problems and their solutions
- CO2: Apply modern engineering tools to attempt solutions to the problems
- CO3: Adapt with computational thinking and automation skills
- CO4: Realize the need of professional communications, team work and self-learning

General Guidelines:

1. Students shall form a group of 3 to 4 students.
2. Each group shall conduct field survey or literature survey in consultation with faculty supervisor/mentor or internal project committee of faculty members to identify the needs of the society.
3. Each group shall formulate the problems on the basis of their survey and submit the implementation plan to the concerned faculty supervisor/mentor or the committee.
4. A log book is to be maintained by each group, wherein the group can record the work progress. The supervisor shall monitor the progress.
5. The faculty supervisor/mentor or the committee may give inputs to the students during the project activities, however, focus shall be on self-learning.
6. Students in a group shall understand problem effectively, propose multiple solutions and select the best solution in consultation with faculty supervisor/mentor or the committee.
7. Students shall convert the best solution into a working model/prototype using various components of their domain areas.
8. Students shall demonstrate and validate the model with proper justification and a report is to be compiled in a standard format/template

Assessment Criteria:

The Micro Project shall be assessed based on the following criteria:

Sl No.	Component	Weightage
1	Proper identification of the problem	20
2	Quality of survey	15
3	Clarity of problem definition	10
4	Presence of innovation	5
5	Effective use of modern engineering tools	10
6	Cost effectiveness	5
7	Societal impact	10
8	Functioning of the working model	15
9	Fulfilment of standard engineering norms	5
10	Clarity in written and oral communication	5
		100

Examinations:

Sl No.	Type of examination/evaluation	Marks (50)
1	Mid-term presentation on Progress	10
2	Demonstration of the project	10
3	End-term presentation	25
4	Viva-voce examination	5
